

UNDERSTANDING THE INDUCTOR: FUNDAMENTALS & APPLICATIONS

1 WHAT IS AN INDUCTOR?

A PASSIVE ELECTRONIC COMPONENT THAT STORES ENERGY IN A MAGNETIC FIELD.



3 KEY FORMULAS & UNIT

MAIN TYPES OF INDUCTORS

Core Material



Physical forms



START HERE



MAGNETIC FIELD

2 HOW IT WORKS (INDUCTION)

- (A) **CONSTANT CURRENT**
(Steady B-field)
- (B) **CHANGING CURRENT**
(Induced B-field opposes change, Lenz's Law)
- (C) **GRAPH OF CURRENT vs. TIME**

CHANGE IN CURRENT

$$\Phi = L \cdot I$$

$$W = \frac{1}{2} \cdot L \cdot I^2$$

HENRY (H)

- Unit: millihenry (mH)
- Unit: microhenry (μH)

COMMON APPLICATIONS



IMPORTANT CHARACTERISTICS

- **DCR (DC Resistance)**
DC Resistance
- **Saturation Current**
Saturation current in gauges
- **Quality Factor (Q)**
Quality factor in even gauges
- **Self-Resonant Frequency (SRF)**